# CO3117 - Part I Summary

Nguyen Thien Nhan - 2452879 - HK261

## 1. Problem and dataset

Use case: predict a person's current physical activity (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) from smartphone accelerometer/gyroscope features. Dataset: UCI Human Activity Recognition Using Smartphones, original 2012 release (DOI 10.24432/C54S4K), frozen in protocol.yaml together with the archive SHA-256. Target: the 6-class activity label. Primary metric: Macro-F1 (accuracy and confusion matrix as secondary evidence). Leakage control: the UCI-provided train/test split is subject-disjoint (21 train subjects, 9 test subjects, no overlap, verified in src/data.py tests); the sealed test population is never touched before Part II; every validation/tuning split in this report is also subject-aware (GroupShuffleSplit on subject_train.txt), never a random row split.

## 2. Representations and protocol

All models in this report use the same frozen 561-feature representation (UCI's precomputed time/frequency-domain features), no additional feature engineering. Every experiment uses the same fit/validation split (GroupShuffleSplit, test_size=0.25, random_state=2452879) carved out of the training population only. Seed 2452879 (student ID) is used consistently across all experiments per protocol.yaml. Environment: Python 3.13, scikit-learn, numpy, see requirements.txt; each experiment script under experiments/part1_pre_midterm/ is self-contained and reproducible by running it directly.

## 3. Model-family map (Part I)

| Model | Ch. | Depth | Code / own implementation |
|---|---|---|---|
| Perceptron (+ one-vs-rest) | 3 | A | src/from_scratch/perceptron.py |
| MLP (forward+backward) | 3 | A | src/from_scratch/mlp.py |
| Decision Tree split routine | 2 | B | src/from_scratch/tree_split.py |
| Naive Bayes (Gaussian+Multinomial) | 4 | A | src/from_scratch/naive_bayes.py |
| Genetic Algorithm (feature selection) | 5 | B | src/from_scratch/genetic_algorithm.py |
| Bayesian Network / TAN (theory) | 6 (Part I) | C | docs/weekly/w07-genetic-algorithm.md |

All Depth-A components pass their contract tests (tests/contract/) and match a trusted reference (sklearn) to tight numerical tolerance where applicable (Perceptron convergence test, MLP numerical-gradient check < 1e-5 relative error, Naive Bayes predict_log_proba matched to sklearn within 1e-4 to 1e-6). MLP is gradient-verified on synthetic data via the contract test but has not yet been trained end-to-end on HAR in this Part I window - noted as a known limitation below rather than claimed as benchmarked.

## 4. Results

Master comparison on the frozen validation split (subject-aware, seed 2452879), Macro-F1 unless noted:

| Model | Macro-F1 | Accuracy | Notes |
|---|---|---|---|
| Perceptron (own, one-vs-rest) | 0.9633 | 0.9636 | linear baseline |
| Perceptron (sklearn) | 0.9734 | 0.9738 | reference |
| Decision Tree (sklearn, depth=8) | 0.858 | - | catchup benchmark, best depth found |
| Naive Bayes (own, var_smoothing=1e-9) | 0.7679 | 0.7752 | matches sklearn 100% |
| Naive Bayes (own, var_smoothing=1e-3) | 0.8411 | - | tuned, best of sweep |
| GA + NB feature selection (3 seeds) | 0.90-0.92 | - | 185-212/561 features, fitness incl. penalty |

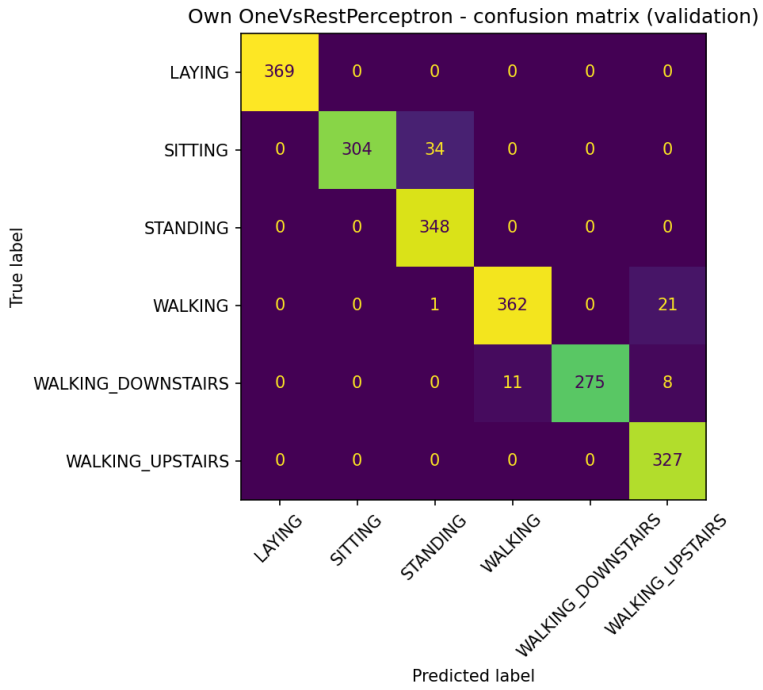



*Figure 1. Own Perceptron confusion matrix (validation).*

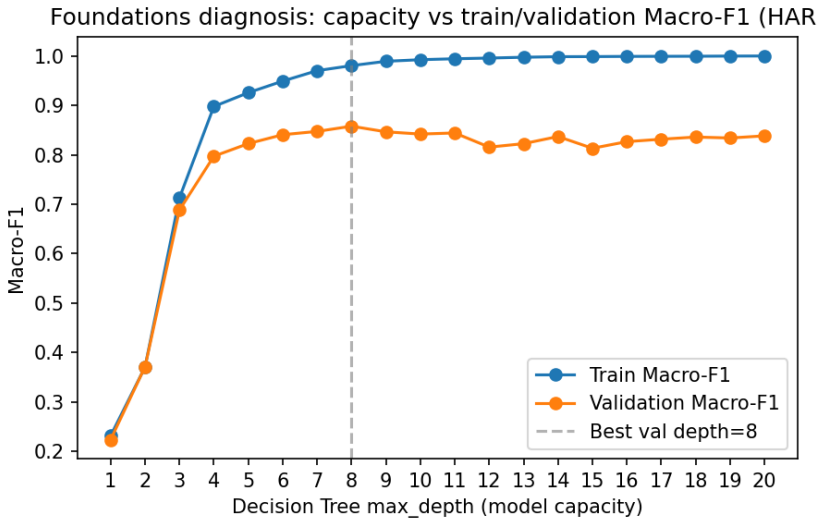



*Figure 2. Decision Tree capacity vs. train/validation Macro-F1 (Ch.1 diagnostic).*

## 5. Error and behavior analysis

The Perceptron's 75/2060 validation errors (3.6%) concentrate on SITTING predicted as STANDING (34/75, ~45%). Naive Bayes shows the same pair but reversed and far worse: 463/2060 errors (22.5%),

with STANDING predicted as SITTING accounting for 169/463 (~37%). Both models struggle hardest on the same pair of static postures, consistent with those two activities producing the most similar accelerometer/gyroscope signatures in this feature space - but Naive Bayes is hurt far more by it, because its conditional-independence assumption double-counts the many correlated features (e.g. multiple axes of the same sensor) that make SITTING and STANDING hard to tell apart, while the Perceptron's single linear boundary per class has no such assumption to violate. The Decision Tree's own capacity sweep shows the classic underfitting (shallow tree, both scores low) to overfitting (max_depth=20, train Macro-F1 to 1.0, validation plateaus/falls) transition, with the best validation score at max_depth=8 in the range tested.

## 6. Cross-model synthesis (Part I)

Generative vs. discriminative is the clearest split in this Part I evidence: the discriminative Perceptron (no independence assumption, just a linear decision boundary) dominates the generative Naive Bayes (assumes conditional feature independence given the class) by a wide margin on this 561-correlated-feature representation. Interestingly, Depth-B feature selection (GA) nearly closes that gap for Naive Bayes - removing redundant/correlated features raises its fitness from 0.768 (all features) to 0.90-0.92 (185-212 selected features), suggesting the independence-violation failure mode is substantially driven by specific redundant feature groups rather than being an inherent ceiling for this dataset. The Decision Tree sits between the two: non-parametric and assumption-light like the Perceptron, but prone to the same kind of overfitting-with-capacity curve classic to Chapter 1's bias-variance discussion.

## 7. Ownership and learning reflection

Concrete ownership evidence from this Part I window: a real ReLU gradient-check failure in the MLP (W05) traced to a zero-bias-init + dead-ReLU interaction, fixed by switching to tanh; a Perceptron-rule vs. Delta-rule misconception corrected by reading ML-From-Scratch's reference code line by line; a GA mutation bug (`self.mutation_rate` vs. the resolved `self.mutation_rate_`) found only once the code actually ran against real data; and a genuine infrastructure failure (GA search crashing from low system memory) that was debugged and made resumable rather than just retried blindly. All Depth-A components were written and sanity-tested before any reference code or AI tool was consulted, per AI_USE.md and the two-state commit history visible in PROGRESS.md. Remaining uncertainty going into Part II: whether the GA's selected feature subsets are converging toward a similar subset across seeds or arriving at similar fitness via different subsets (not yet checked), and whether MLP would close more of the gap to the Perceptron once actually trained on HAR.

## 8. Reproducibility and references

Setup: python -m venv .venv; pip install -r requirements.txt; python tools/check.py. Every experiment script under experiments/part1_pre_midterm/ is runnable directly and reads the frozen split/seed from protocol.yaml / src/data.py. Full citations in REFERENCES.md; AI use disclosed per-episode in AI_USE.md (Socratic/verification/debugging use only, after each first-attempt commit, per assignment Section 12).